

ORIGINAL RESEARCH

Treatment of Cardiometabolic Risk Factors Among U.S. Adults With Cardiovascular-Kidney-Metabolic Syndrome



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ABSTRACT

BACKGROUND Cardiovascular-kidney-metabolic (CKM) syndrome highlights the inter-related nature of cardiometabolic risk factors, kidney disease, and cardiovascular disease (CVD) and represents an important target for intervention amid rising cardiovascular mortality in the United States. However, contemporary treatment patterns of major cardiometabolic risk factors in this high-risk population are not well defined.

OBJECTIVES The purpose of this study was to characterize national treatment rates of hypertension, diabetes, and hyperlipidemia among U.S. adults with CKM syndrome and to assess risk factor control among treated individuals from 2015 through 2023.

METHODS We analyzed data from adults aged ≥ 20 years with CKM stage 2 and above who participated in the National Health and Nutrition Examination Survey from 2015 through August 2023. Treatment rates for hypertension, diabetes, and hyperlipidemia and rates of blood pressure, glycemic, and cholesterol control among treated individuals were estimated using age- and sex-adjusted analyses that accounted for the complex survey design.

RESULTS The study included 6,384 adults with CKM stage 2 and above (weighted mean age 44.2 years; 51.2% women). Only approximately one-half of adults with hypertension (51.3% [95% CI: 49.7%-52.8%]) or hyperlipidemia (48.8% [95% CI: 46.7%-51.0%]) were receiving treatment, while diabetes treatment rates were higher (83.4% [95% CI: 80.1%-86.6%]). Among treated individuals, blood pressure control was achieved in 44.7% (95% CI: 41.6%-47.7%), glycemic control in 47.3% (95% CI: 42.7%-51.8%), and cholesterol control in 68.2% (95% CI: 65.2%-71.1%). Treatment rates for hypertension and hyperlipidemia increased across higher risk strata, while blood pressure and glycemic control were lower among treated adults with higher 10-year CVD risk. Adults aged 20-44 years consistently had the lowest treatment rates across all 3 conditions (hypertension 27.6% [95% CI: 24.8%-30.3%], diabetes 73.7% [95% CI: 63.8%-83.7%], hyperlipidemia 19.7% [95% CI: 15.2%-24.3%]), while women were less likely than men to receive treatment for diabetes and hyperlipidemia. Among racial and ethnic subgroups, Hispanic adults had the lowest treatment rates for hypertension and hyperlipidemia.

CONCLUSIONS Among U.S. adults with CKM syndrome, treatment of hypertension and hyperlipidemia was low, and fewer than one-half of treated individuals achieved blood pressure or glycemic control. Gaps in treatment initiation were most pronounced among young adults, women, and Hispanic adults, and inadequate risk factor control was particularly evident among those with higher cardiovascular risk. These findings highlight substantial opportunities to improve cardiometabolic care in this high-risk population. (JACC. 2026;88:613-624) © 2026 by the American College of Cardiology Foundation.

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**ABBREVIATIONS
AND ACRONYMS****CKM** = cardiovascular-kidney-metabolic**CVD** = cardiovascular disease

The complex interplay among cardiometabolic risk factors, renal health, and cardiovascular disease (CVD) has gained increasing recognition. The American Heart Association (AHA) recently introduced the framework of cardiovascular-kidney-metabolic (CKM) health to highlight the interconnectedness of these conditions and to promote integrated approaches to clinical care.¹ As the United States faces deteriorating cardiometabolic health and rising cardiovascular mortality, population CKM health has become a pressing public health imperative.²⁻⁷

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Central to improving CKM health outcomes is the effective management of cardiometabolic risk factors, which not only define early CKM stages but also drive progression across the CKM continuum and contribute directly to adverse events. Despite the availability of an expanding portfolio of therapies that act across multiple CKM pathways, control of risk factors remains suboptimal, particularly among individuals with multiple CKM conditions.⁸⁻¹⁰ Treatment initiation is the first critical step toward achieving adequate risk factor control; yet, little is known about the extent to which therapies are initiated for major cardiometabolic risk factors in the CKM population—both overall and by CKM stage.

In this study, we aimed to answer 3 questions. First, what were treatment rates for hypertension, diabetes, and hyperlipidemia among U.S. adults with CKM syndrome between 2015 and 2023? Second, among those who received treatment, what proportion achieved control of key cardiometabolic risk factors? Third, how did the treatment and control rates vary across cardiovascular risk strata and sociodemographic subgroups?

METHODS

DATA SOURCE. We used data from the 2015-2017 to 2021-August 2023 cycles of the National Health and Nutrition Examination Survey (NHANES). NHANES is a nationally representative survey that uses a complex, multistage probability sampling design to monitor the health status of the noninstitutionalized U.S. population. The survey is conducted by the Centers for Disease Control and Prevention in 2-year

cycles and is administered by trained personnel through in-person interviews and medical examinations, which include physical examinations and laboratory measurements.

STUDY POPULATION. We identified nonpregnant adults aged ≥ 20 years with at least stage 2 CKM in NHANES. CKM stages are defined based on the 2023 AHA Presidential Advisory on CKM Health.^{1,11} Those with CKM stage 0 or 1 were not included as they by definition did not have cardiometabolic risk factors of interest (hypertension, diabetes, hyperlipidemia).

To capture those with stage 2 CKM and above, we included participants with ≥ 1 cardiometabolic risk factors or moderate-to-very-high-risk CKD per Kidney Disease Improving Global Outcomes criteria, or established CVD consistent with AHA recommendations.¹ Qualifying cardiometabolic risk factors included fasting serum triglycerides ≥ 135 mg/dL, hypertension, diabetes, or metabolic syndrome. Hypertension was defined as a systolic blood pressure ≥ 130 mm Hg, a diastolic blood pressure ≥ 80 mm Hg, or a reported health care diagnosis.^{12,13} Blood pressure was calculated as the mean of 3 measurements taken after the participant had rested quietly for 5 minutes. Diabetes was defined as a glycosylated hemoglobin $\geq 6.5\%$, fasting plasma glucose level ≥ 126 mg/dL (6.99 mmol/L), or a health care diagnosis of diabetes.² Hyperlipidemia was defined as total cholesterol ≥ 200 mg/dL or a self-reported health care diagnosis of hyperlipidemia, consistent with recent AHA and Centers for Disease Control and Prevention surveillance reports and prior studies.^{3,14-16}

Metabolic syndrome was defined as ≥ 3 of the following: 1) elevated waist circumference (≥ 90 cm for Asian men, ≥ 80 cm for Asian women, ≥ 102 cm for men, and ≥ 88 cm for women from all other racial ethnic groups); 2) low high-density lipoprotein levels (< 40 mg/dL for men, < 50 mg/dL for women); 3) fasting serum triglycerides ≥ 150 mg/dL; 4) elevated blood pressure (systolic blood pressure ≥ 130 mm Hg, diastolic blood pressure ≥ 80 mm Hg, and/or use of blood pressure-lowering medications); and 5) fasting glucose ≥ 100 mg/dL. CKD stage was defined using estimated glomerular filtration rate (eGFR) and urinary albumin-creatinine-ratio. eGFR was calculated using the race-free CKD-EPI 2021 creatinine equation.¹ Moderate-risk CKD includes those with CKD stage G1 or G2 with albuminuria stage A2, CKD

The authors attest they are in compliance with human studies committees and animal welfare regulations of the authors' institutions and Food and Drug Administration guidelines, including patient consent where appropriate. For more information, visit the [Author Center](#).

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stage G3a with albuminuria stage A1. High-risk CKD includes those who have CKD stage G1 or G2 with albuminuria stage A3, CKD stage G3a with albuminuria stage A2, or CKD stage G3b with albuminuria stage A1. Very high-risk CKD includes those with CKD stage G3a with albuminuria stage A3, CKD stage G3b with albuminuria stage A2 and A3, CKD stage G4, or CKD stage G5. Self-reported CVD included coronary heart disease, myocardial infarction, angina, stroke, or heart failure. Atrial fibrillation and peripheral artery disease were not included because NHANES does not routinely ascertain these conditions.

RISK STRATIFICATION OF PARTICIPANTS. To assess treatment rates of key cardiometabolic conditions across risk strata, participants with CKM stage 2 and above were stratified by 10-year CVD risk using the AHA PREVENT (Predicting Risk of CVD EVENTS) equations.¹⁷ Four groups were defined: low-to-borderline risk (<7.5%), intermediate risk (7.5% to <20%), high risk ($\geq 20\%$), and those with established CVD. Given that the PREVENT model was developed and validated among those aged 30 to 79 years, risk was not estimated for participants aged <30 years; those ≥ 80 years were assigned an age of 79 years for risk calculation.¹¹ The allowable ranges for cardiometabolic measures required by PREVENT were as follows: total cholesterol 130 to 320 mg/dL; high-density lipoprotein 20 to 100 mg/dL, systolic blood pressure 90 to 200 mm Hg, body mass index 18.5 to 39.9 kg/m², and eGFR 15 to 140 mL/min/1.73 m². For values outside these bounds, measurements were truncated to the nearest limit to enable risk estimation using PREVENT.

OUTCOMES. The main outcomes were as follows: 1) treatment rates for hypertension, diabetes, and hyperlipidemia among adults with CKM stage 2 and above; and 2) blood pressure, glycemic, and cholesterol control among those receiving treatment for these respective conditions.

Treatment of hypertension was assessed among those with hypertension and was defined as answering “yes” to the question “Are you now taking any medication prescribed by a doctor for your high blood pressure?” Treatment of diabetes was assessed among those with diabetes and was defined as answering “yes” to either “Are you now taking diabetic pills to lower blood sugar?” or “Are you now taking insulin?” Treatment of hyperlipidemia was assessed among those with hyperlipidemia and was defined as answering “yes” to “To lower your blood cholesterol, have you ever been told by a doctor or other health professional to take prescribed

medicine?” Blood pressure control was defined as systolic blood pressure <130 mm Hg and diastolic blood pressure <80 mm Hg among adults being treated with an antihypertensive medication.^{12,18} Glycemic control was defined as glycosylated hemoglobin <7%^{19,20} and cholesterol control was defined as total cholesterol <200 mg/dL among those receiving treatment for each respective condition.¹⁵

STATISTICAL ANALYSIS. We first determined baseline characteristics of U.S. adults with CKM stage 2 and above, accounting for the NHANES complex survey design. Next, we calculated the age- and sex-adjusted treatment rates of hypertension, diabetes, and hyperlipidemia using direct standardization to the 2020 U.S. population as a reference. The same approach was used to determine age- and sex-adjusted rates of blood pressure, glycemic, and cholesterol control among those receiving treatment. We then repeated the main analysis to examine outcomes by cardiovascular risk strata and sociodemographic factors, including age, sex, and race and ethnicity. A survey-weighted modified Poisson regression model with a log link was fit to estimate the association between changes in PREVENT-predicted 10-year CVD risk as a continuous variable and outcomes. Finally, we fit survey-weighted linear regression models to evaluate overall temporal changes in treatment and control among treated individuals for each cardiometabolic condition over the study period. All analyses incorporated NHANES survey weights to obtain nationally representative estimates.

Additional analyses. We performed several additional analyses. First, hypertension treatment and blood pressure control were reassessed using a 140/90 mm Hg threshold, reflecting older guidelines that were in place during the first 2 years of our study period.¹³ Second, because hyperlipidemia treatment recommendations for low- and borderline-risk individuals depend on risk level and the presence of risk enhancers, the absence of treatment initiation in these groups may represent guideline-concordant care.²¹ We therefore examined hyperlipidemia treatment rates among adults with diabetes, for whom statin therapy is recommended after age 40 years irrespective of predicted risk, to differentiate true treatment gaps from appropriate nontreatment. Third, to understand how reliance on self-reported diagnosis for cohort inclusion might bias treatment rate estimates, we calculated the proportion of participants qualifying for each cardiometabolic risk factor by self-reported history alone, elevated biometric and/or laboratory measures alone, or both,

TABLE 1 Baseline Characteristics of U.S. Adults With Cardiovascular-Kidney-Metabolic Syndrome (Stage 2 and Above), 2015 to 2023^a

	NHANES Cycles, % (95% CI)			
	Overall	2015-2016 (n = 1,594)	2017 to March 2020 (n = 2,709)	2021-2023 (n = 2,081)
Mean age, y	44.2 (43.4-44.9)	43.7 (42.4-45.0)	44.2 (42.9-45.5)	44.6 (43.3-45.9)
Sex				
Men	48.8 (47.7-49.9)	48.5 (46.8-50.2)	48.7 (46.5-50.9)	49.3 (47.5-51.1)
Women	51.2 (50.1-52.3)	51.5 (49.8-53.2)	51.3 (49.1-53.5)	50.7 (48.9-52.5)
Race and ethnicity ^b				
Non-Hispanic White	60.5 (57.1-63.8)	62.5 (54.1-70.8)	60.6 (56.5-64.6)	58.5 (54.5-62.5)
Hispanic	17.1 (14.2-20.0)	16.0 (10.2-21.8)	17.4 (14.0-20.8)	17.9 (12.4-23.4)
Non-Hispanic Black	11.5 (9.4-13.6)	11.7 (6.8-16.6)	11.7 (8.7-14.7)	11.1 (8.2-13.9)
Asian/other	10.9 (9.5-12.4)	9.8 (7.8-11.8)	10.4 (8.1-12.6)	12.6 (9.5-15.7)
Family income-to-poverty ratio				
<200%	34.1 (31.2-36.9)	36.3 (30.6-41.9)	33.5 (30.6-36.4)	32.4 (26.6-38.3)
≥200%	65.9 (63.1-68.8)	63.7 (58.1-69.4)	66.5 (63.6-69.4)	67.6 (61.7-73.4)
Cardiovascular risk strata ^c				
Low-to-borderline risk (<7.5%)	58.1 (55.9-60.1)	56.2 (52.2-60.3)	56.6 (52.4-60.7)	60.8 (58.2-63.4)
Intermediate risk (7.5% to <20%)	21.7 (20.3-23.1)	23.8 (21.1-26.5)	20.9 (18.3-23.6)	20.5 (18.8-22.2)
High risk (20% and greater)	6.8 (5.9-7.6)	6.8 (5.2-8.3)	6.9 (5.5-8.3)	6.6 (5.3-7.9)
History of cardiovascular disease	13.5 (12.3-14.7)	13.2 (11.2-15.1)	15.5 (12.9-18.2)	12.1 (10.6-13.5)

^aNationally representative estimates of U.S. adults aged 20 and older from National Health and Nutrition Examination Survey (NHANES), 2015 to 2023. Age, sex, race/ethnicity, and family income are based on self-report. ^bRace and ethnicity were based on self-report in closed categories. Asian, Black, and White adults consisted of individuals self-reporting as non-Hispanic. Hispanic individuals included all adults self-reporting as other Hispanic and Mexican American. Black, Mexican American, and other Hispanic were oversampled in all survey cycles. ^cCardiovascular risk strata were defined based on 10-year risk for cardiovascular disease defined by PREVENT (Predicting Risk of CVD EVENTS) calculator and prior history of cardiovascular disease (self-reported history of coronary heart disease, myocardial infarction, angina, stroke, or heart failure).

and then compared treatment rates across these subgroups. Fourth, to assess whether gaps in treatment reflected appropriate surveillance in lower-risk individuals or potential missed opportunities, we compared cardiometabolic measures across CV risk strata for the entire cohort and the untreated group. Fifth, to determine the potential influence of access to care on outcomes, we examined insurance status, presence of usual source of care, and type of usual source of care by treatment status (eg, receipt of treatment for qualifying cardiometabolic conditions) (Supplemental Methods). Finally, participants were categorized by the number of cardiometabolic conditions (hypertension, diabetes, hyperlipidemia) to characterize overall disease burden, and outcomes of interest were assessed across these categories.

Analyses were conducted using R version 4.4.2 (R Foundation). Statistical significance was defined as 2-sided $P < 0.05$. The NHANES protocols were approved by the National Center for Health Statistics and participants provided written informed consent.

RESULTS

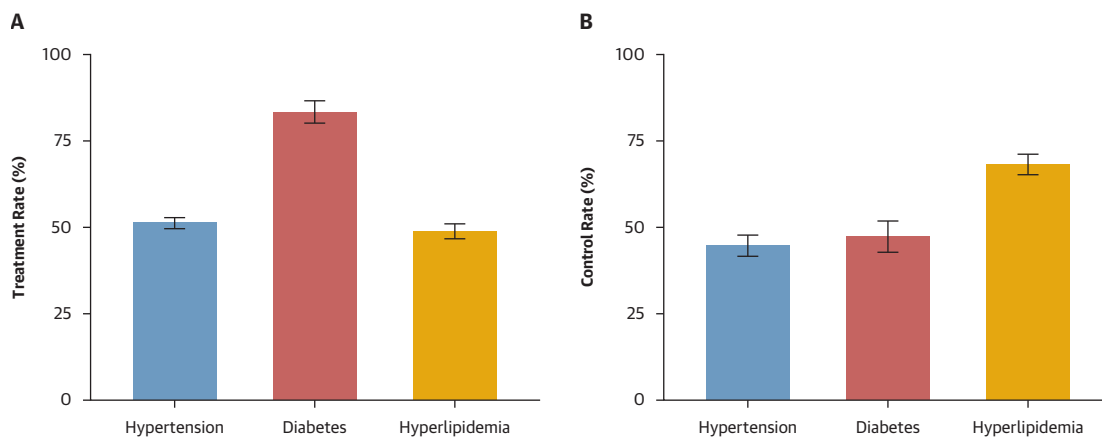
The study population included 6,384 adults with CKM stage 2 and above (weighted mean age 44.2 years; 51.2% women). Of this population, 1,098 had prior CVD and 4,006 were eligible for 10-year CVD

risk estimation by PREVENT. Baseline characteristics are shown in Table 1.

TREATMENT AND CONTROL OF CARDIOMETABOLIC RISK FACTORS. Among adults with CKM stage 2 and above, 71.5% (95% CI: 69.8%-73.2%) had hypertension, 21.8% (95% CI: 20.4%-23.3%) had diabetes, and 58.9% (95% CI: 57.1%-60.8%) had hyperlipidemia. Only 51.3% (95% CI: 49.7%-52.8%) of those with hypertension were receiving treatment (Figure 1A), and less than one-half of treated individuals achieved blood pressure control (44.7% [95% CI: 41.6%-47.7%]) (Figure 1B). Treatment rates for hypertension decreased over the study period (55.4% [95% CI: 52.1%-58.7%] to 43.9% [95% CI: 41.1%-46.7%]) while blood pressure control among treated individuals were unchanged (44.7% [95% CI: 38.4%-51.1%] to 45.5% [95% CI: 40.7%-50.3%]) (Figure 2).

In contrast to hypertension, treatment rates for diabetes were higher (83.4% [95% CI: 80.1%-86.6%]) (Figure 1A), although glycemic control rates remained low (47.3% [95% CI: 42.7%-51.8%]) (Figure 1B). Although treatment rates for diabetes increased over the study period (75.9% [95% CI: 70.4%-81.4%] to 88.0% [95% CI: 82.2%-93.7%]), glycemic control remained unchanged among treated individuals (48.4% [95% CI: 38.6%-58.3%] to 42.4% [95% CI: 38.1%-46.8%]) (Figure 2).

FIGURE 1 Treatment and Control Among Adults With Cardiovascular-Kidney-Metabolic Syndrome (Stage 2 and Above), 2015 to 2023



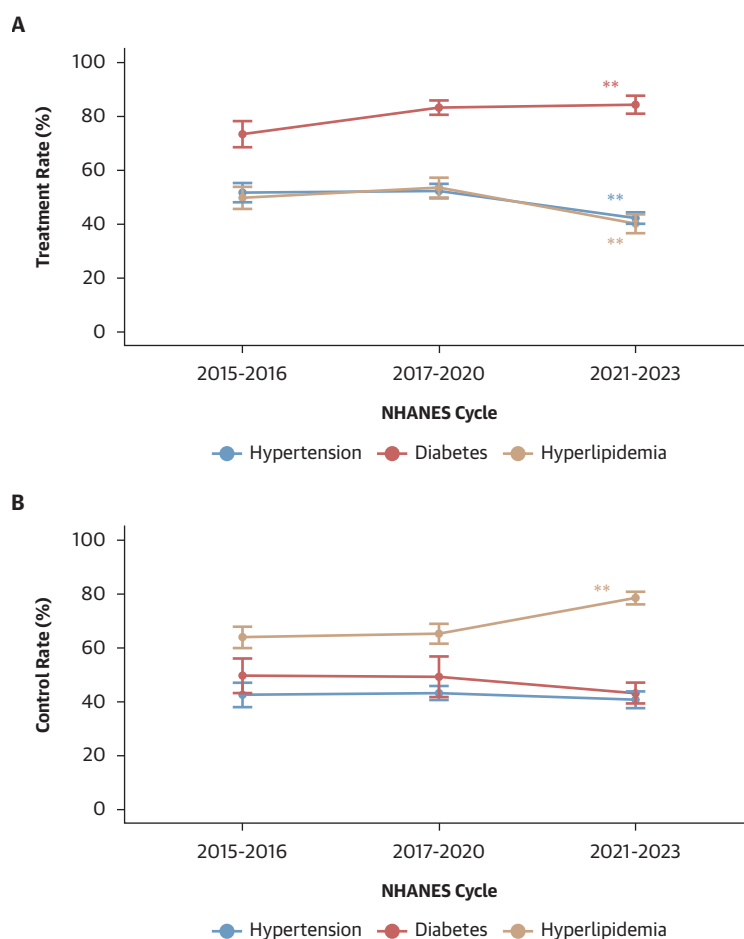
(A) Treatment of cardiometabolic risk factors. Treatment of hypertension (blue), diabetes (red), and hyperlipidemia (yellow) among U.S. adults with cardiovascular-kidney-metabolic syndrome (stage 2 and above) based on the National Health and Nutrition Examination Survey, 2015-2016 to 2021-2023. (B) Control of cardiometabolic risk factors among treated individuals. Blood pressure (blue), glycemic (red), and cholesterol (yellow) control among those with cardiovascular-kidney-metabolic syndrome (stage 2 and above) who were treated for hypertension, diabetes and hyperlipidemia based on the National Health and Nutrition Examination Survey, 2015-2016 to 2021-2023. Whiskers represent 95% CIs.

Among adults with hyperlipidemia, only 48.8% (95% CI: 46.7%-51.0%) were receiving treatment, and 68.2% (95% CI: 65.2%-71.1%) achieved control. Notably, hyperlipidemia treatment rates decreased from 53.0% (95% CI: 48.3%-57.7%) to 39.8% (95% CI: 36.6%-43.1%) from 2015-2016 to 2021-2023 while cholesterol control increased (64.7% [95% CI: 59.2%-70.3%] to 76.5% [95% CI: 72.1%-80.9%]) (Figure 2).

CARDIOVASCULAR RISK STRATA. Hypertension treatment was lowest among adults with low-to-borderline CVD risk (38.4% [95% CI: 35.4%-41.4%]) (Figure 3A), but higher among those with intermediate CVD risk (68.4% [95% CI: 64.5%-72.3%]), high CVD risk (77.5% [95% CI: 73.2%-81.8%]), and established CVD (80.1% [95% CI: 75.9%-84.2%]). Among adults without established CVD, hypertension treatment rates increased by 18% for every 5% increase in estimated 10-year CVD risk (relative risk [RR]: 1.18 [95% CI: 1.16-1.20]) (Supplemental Table 1). Among individuals receiving antihypertensive treatment, fewer than one-half achieved blood pressure control across all risk strata (low-to-borderline CVD risk: 46.5% [95% CI: 40.3%-52.8%]; intermediate CVD risk: 48.2% [95% CI: 43.4%-53.1%]; high CVD risk: 34.3% [95% CI: 29.3%-39.4%]; established CVD: 41.7% [95% CI: 36.5%-46.8%]) (Figure 3B). Blood pressure control declined by ~4% (RR: 0.96 [95% CI: 0.92-0.99]) with every 5% increase in 10-year CVD risk (Supplemental Table 1).

Diabetes treatment rates were consistently high across cardiovascular risk strata (low-to-borderline CVD risk: 82.4% [95% CI: 74.9%-89.8%]; intermediate CVD risk: 84.9% [95% CI: 80.4%-89.3%]; high CVD risk: 85.1% [95% CI: 79.3%-90.8%]; established CVD: 83.5% [95% CI: 79.4%-87.7%]) (Figure 3A). Despite high treatment rates, glycemic control among treated individuals was suboptimal across all risk strata (low-to-borderline CVD risk: 55.5% [95% CI: 44.5%-66.6%]; intermediate CVD risk: 49.3% [95% CI: 40.9%-57.7%]; high CVD risk: 52.7% [95% CI: 42.5%-62.9%]; established CVD: 40.8% [95% CI: 33.4%-48.3%]) (Figure 3B). There was no significant association between PREVENT estimated 10-year CVD risk and either treatment or glycemic control (Supplemental Table 1).

Hyperlipidemia treatment rates were low among adults with low-to-borderline CVD risk (30.4% [95% CI: 26.5%-34.2%]), but higher among those with intermediate CVD risk (63.3% [95% CI: 58.9%-67.6%]), high CVD risk (75.1% [95% CI: 69.7%-80.5%]), and established CVD (80.3% [95% CI: 75.8%-84.8%]) (Figure 3A). Among those who received treatment, cholesterol control increased with increasing estimated CVD risk (low-to-borderline CVD risk, 53.4% [95% CI: 45.3%-61.5%]; intermediate CVD risk, 65.9% [95% CI: 60.8%-71.1%]; high CVD risk, 75.8% [95% CI: 69.9%-81.6%]; established CVD, 80.9% [95% CI: 77.0%-84.9%]) (Figure 3B, Supplemental Table 1).

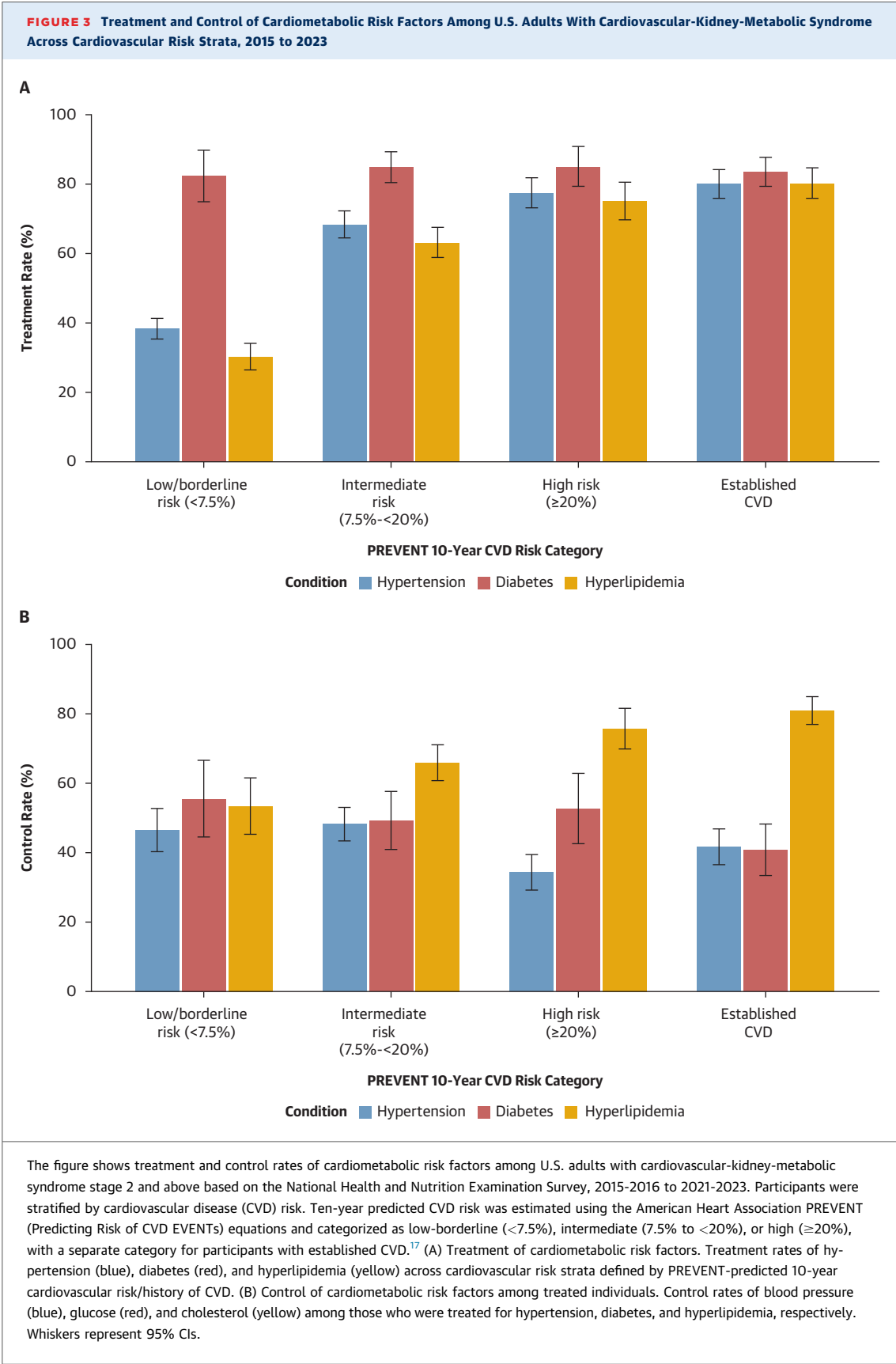
FIGURE 2 Temporal Trends in Treatment and Risk Factor Control Among U.S. Adults With Cardiovascular-Kidney-Metabolic Syndrome, 2015 to 2023

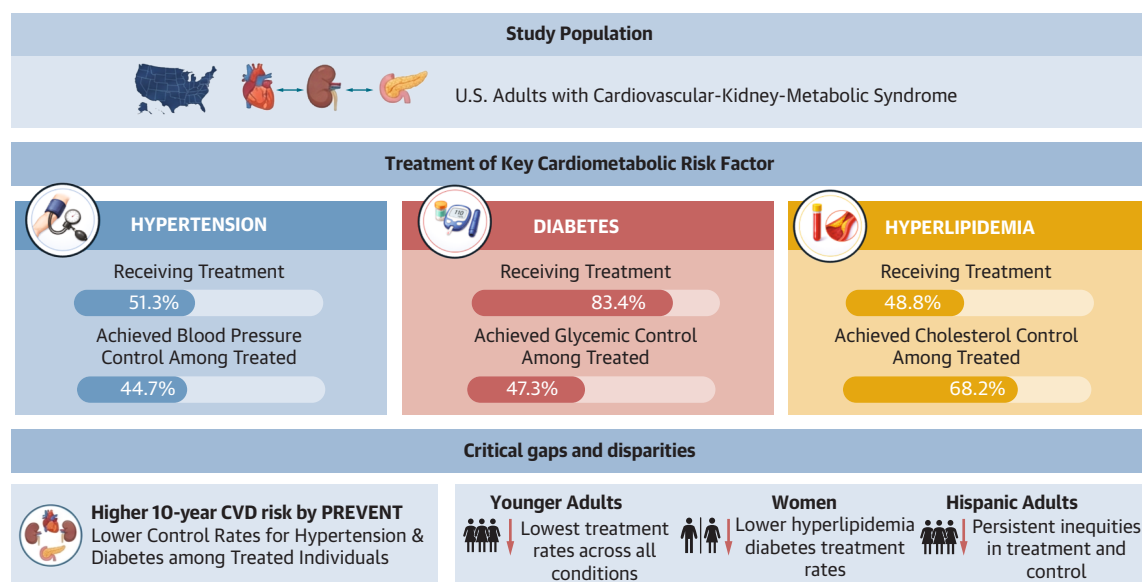
Survey-weighted treatment rates of hypertension, diabetes, hyperlipidemia, and control rates of corresponding cardiometabolic measures in treated individuals, among U.S. adults with at least stage 2 cardiovascular-kidney-metabolic syndrome from the National Health and Nutrition Examination Survey (NHANES), 2015-2016 to 2021-2023. Error bars indicate 95% CIs. $^{**}P < 0.01$. (A) Temporal trends in treatment of cardiometabolic risk factors. Survey-weighted proportion of U.S. adults with at least stage 2 cardiovascular-kidney-metabolic syndrome treated for hypertension (blue), diabetes (red), and hyperlipidemia (yellow). (B) Temporal trends in control of cardiometabolic risk factors among treated individuals. Survey-weighted proportion of U.S. adults with blood pressure control, glycemic control, and cholesterol control among individuals with stage 2 cardiovascular-kidney-metabolic syndrome and were treated for hypertension (blue), diabetes (red), and hyperlipidemia (yellow).

SOCIODEMOGRAPHIC SUBGROUPS. Across age subgroups, treatment rates were consistently lowest among younger adults aged 20-44 years for all 3 conditions—hypertension (27.6% [95% CI: 24.8%-30.3%]), diabetes (73.7% [95% CI: 63.8%-83.7%]), and hyperlipidemia (19.7% [95% CI: 15.2%-24.3%])—compared with middle-aged and older adults. Women were more likely than men to receive treatment for hypertension (55.7% [95% CI: 53.1%-58.4%] vs 47.5% [95% CI: 45.5%-49.5%]), but were less likely to receive treatment for diabetes (77.3% [95% CI: 72.4%-82.2%] vs 87.8% [95% CI: 84.1%-91.6%]) or

hyperlipidemia (45.9% [95% CI: 42.4%-49.5%] vs 51.4% [95% CI: 48.0%-54.7%]) (Supplemental Tables 2 to 4). Rates of risk factor control varied across age and sex subgroups and are shown in Supplemental Tables 2 to 4.

Across racial and ethnic subgroups, Hispanic adults were less likely to receive treatment for hypertension (44.6% [95% CI: 40.3%-49.0%]) compared with non-Hispanic White adults (52.6% [95% CI: 50.0%-55.1%]) (Supplemental Table 2). Among treated individuals, blood pressure control was substantially worse among Black individuals compared



CENTRAL ILLUSTRATION Cardiometabolic Risk Factor Treatment and Control Among U.S. Adults With Cardiovascular-Kidney-Metabolic Syndrome

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Treatment rates of cardiometabolic risk factors, as well as control among those receiving treatment, among U.S. adults with cardiovascular-kidney-metabolic syndrome in National Health and Nutrition Examination Survey. CVD = cardiovascular disease.

with non-Hispanic White adults (34.8% [95% CI: 30.8%-38.7%] vs 46.8% [95% CI: 42.6%-51.0%]) (Supplemental Table 2). For diabetes, Hispanic adults, when treated, were less likely to achieve glycemic control (37.8% [95% CI: 31.2%-44.5%] vs 52.6% [95% CI: 44.4%-60.9%]) compared with non-Hispanic White adults (Supplemental Table 3). Cholesterol treatment and control rates also varied across racial and ethnic groups and are shown in Supplemental Table 4.

ADDITIONAL ANALYSES. First, using a more lenient blood pressure threshold (140/90 mm Hg) based on older guidelines, 68.6% (95% CI: 66.8%-70.4%) of adults with CKM and hypertension received treatment, and 70.5% (95% CI: 68.8%-72.2%) of treated individuals achieved blood pressure control. Second, among participants with diabetes, treatment rates of hyperlipidemia were lowest in those with low-to-borderline CVD risk compared with those with higher CVD risk or established CVD, mirroring patterns observed across cardiovascular risk strata in the overall CKM cohort (Supplemental Figure 1). Third, treatment rates were significantly higher for those with cardiometabolic risk factors defined by self-report alone or self-report in addition to elevated biometric and/or laboratory measures compared with

those with elevated biometric and/or laboratory values alone (Supplemental Table 5). Fourth, among those not receiving treatment for hypertension or hyperlipidemia, disease severity was comparable to or worse than that of the overall CKM population with these conditions (Supplemental Table 6). In contrast, those with diabetes not receiving treatment had less severe disease compared with all participants with diabetes. Fifth, to provide mechanistic insight into the observed treatment gaps, we examined indicators of health care contact. Participants who did not receive treatment for any qualifying cardiometabolic conditions were more likely to be uninsured and to lack a usual source of care compared with those who received treatment (Supplemental Table 7). Finally, individuals in higher CVD risk strata had a greater burden of cardiometabolic conditions (Supplemental Table 8); treatment gaps narrowed as the number of conditions increased, while control among treated individuals was similar across risk strata (Supplemental Figure 2, Supplemental Table 9).

DISCUSSION

In a national study of U.S. adults with CKM syndrome, we identified substantial gaps in the

treatment and control of cardiometabolic risk factors. Only about one-half of adults with hypertension or hyperlipidemia were receiving treatment, compared with substantially higher treatment rates for diabetes, with the most striking gaps observed among younger adults, women, and Hispanic adults. However, even among treated individuals, fewer than one-half achieved blood pressure and glycemic control, underscoring major shortfalls in risk factor management that extend well beyond treatment initiation. Inadequate blood pressure and glycemic control was most pronounced among adults at the highest CVD risk (**Central Illustration**). Together, these findings reveal pervasive gaps in cardiometabolic risk management and highlight opportunities to reduce long-term cardiovascular morbidity and mortality by prioritizing younger populations and improving risk factor control among those at highest CVD risk.

Our findings have important public health implications. As the United States confronts worsening cardiometabolic health, population-level CKM health has emerged as an important public health priority.^{2-4,14,22} Effective treatment and sustained control of cardiometabolic risk factors are foundational to this effort. Durable risk factor control has the potential to prevent downstream adverse cardiovascular events and reduce cardiovascular mortality. Our findings build on prior studies—including those examining risk factor control alone¹⁰ and estimating the proportion of U.S. adults eligible for CKM conditions²³—by demonstrating substantial gaps in both treatment initiation and adequate risk factor control among treated individuals across the continuum of cardiovascular risk. The largest gaps were observed among younger adults in treatment initiation and among individuals at higher CVD risk in achieving adequate control—populations in whom earlier, more intensive intervention may yield the greatest population-level gains. Recent national policy initiatives increasingly seek to incentivize improvements in CKM care by aligning clinical reimbursement with evidence-based management of cardiometabolic risk factors,²⁴ reinforcing the central role of frontline clinicians in improving population CKM health.

Hypertension treatment rates in the CKM population were low and declined over the study period, with fewer than one-half of treated patients achieving blood pressure control, while treatment rates for hyperlipidemia were similarly low and continued to decline. The low treatment rate for hypertension in the low-risk group likely reflects the limited evidence-base for pharmacologic

intervention at this stage of risk, underscoring the need for prospective studies to clarify optimal treatment thresholds. More broadly, treatment gaps may also reflect ongoing uncertainty regarding optimal blood pressure targets, particularly among individuals with diabetes.²⁵⁻²⁷ Our finding that blood pressure control was poorest among individuals at highest CVD risk underscore the significant barriers associated with not only treatment initiation but also timely intensification; these include clinical inertia,^{28,29} growing constraints on access to primary care,³⁰ the competing demands of multimorbidity, the complexity of frequent medication titration, and in some cases, the burden of polypharmacy in this population.³¹ Left unaddressed, these widening gaps may amplify long-term cardiovascular morbidity and mortality in an already high-risk population. Achieving population-level improvements in CKM health will therefore require coordinated, multilevel strategies that harness innovative care-delivery models, expand awareness and access, and strengthen clinician screening and treatment practices.

Although treatment rates for diabetes were high, fewer than one-half of treated individuals achieved glycemic control, highlighting persistent challenges in achieving adequate diabetes management at the population level. Our observation that untreated individuals with diabetes had lower glycosylated hemoglobin suggests that, in some cases, deferring pharmacotherapy may reflect lower disease severity and represent a clinically appropriate decision. At the same time, lack of treatment may also reflect clinical inertia in early-stage disease—a missed opportunity, given that earlier intervention has the potential to yield durable long-term benefit. Despite therapeutic advances, worsening lifestyle factors, including increased consumption of ultra-processed foods and rising sedentary behaviors, have likely attenuated the benefits of newer pharmacological agents.^{32,33} Unlike blood pressure treatment, which often produces rapid therapeutic response, improvements in glycemic control typically occur over months, thereby requiring sustained engagement between patients and clinicians and longitudinal follow-up. The persistently low rates of glycemic control may also reflect the lasting influence of landmark trials published in 2008 and 2009, which demonstrated that intensive glycemic control conferred no additional protective benefits against cardiovascular disease.³⁴⁻³⁷ Strategies to improve glycemic control may include earlier initiation of insulin therapy to slow disease progression, more systematic identification and treatment of prediabetes, and broader use

of novel glucose-lowering therapies with more durable effects.³⁸⁻⁴⁰ Realizing these gains will require coordinated action spanning clinical practice, health system design, and public health infrastructure.

The low rates of treatment for young adults with hypertension and hyperlipidemia are particularly striking. Younger adults have fewer interactions with primary care and higher rates of uninsurance, which may limit opportunities for diagnosis and treatment initiation.^{4,41-43} Early-life exposure to elevated cholesterol and other undertreated cardiometabolic risk factors confers lasting cardiovascular risk that may persist despite effective management later in life, with potentially greater consequences among individuals with CKM.^{4,44,45} Furthermore, we observed that women with CKM had higher treatment rates for hypertension but lower treatment rates for hyperlipidemia and diabetes. The sex differences in treatment rates likely reflect disparities in access, implicit bias, and prevailing clinical practice patterns.⁴⁶ For example, women have been shown to be less likely to be offered statin therapy when eligible and, when offered, more likely to decline initiation or discontinue treatment.⁴⁷ We also observed marked racial and ethnic disparities, characterized by worse blood pressure control among Black adults and lower treatment rates for hypertension and hyperlipidemia among Hispanic adults, which likely reflect persistent inequities in insurance coverage, access to care, and health care affordability, often compounded by socioeconomic and language barriers.⁴⁸⁻⁵⁴

Together, our findings underscore the need for targeted, multilevel clinical, public health, and policy efforts to improve treatment initiation and appropriate titration of therapies for key cardiometabolic risk factors, particularly among adults at elevated CVD risk, young adults, women, and Black and Hispanic populations, to mitigate widening disparities in treatment and cardiovascular outcomes among individuals with CKM syndrome. Our study provides novel and contemporary insights into cardiometabolic risk factor management in this population and identifies inadequate treatment initiation as a central mechanism underlying the persistently poor control of cardiometabolic risk factors.¹⁰

STUDY LIMITATIONS. First, the response rate for NHANES declined over the study period. However, NHANES sampling weights account for nonresponse bias and were used in our study. Second, history of CVD and treatment information was obtained by self-report and is subject to misclassification. Specifically, treatment misclassification may have occurred

because NHANES does not capture injectable therapies (eg, GLP-1 receptor agonists), potentially resulting in classification of treated individuals as untreated, and because participants with clinical cardiovascular disease may have been prescribed medications that affect cardiometabolic measures for other indications. Third, our approach likely overestimated the population eligible for treatment, based on total cholesterol level and health care diagnosis alone rather than risk-based framework suggested by guidelines, and underestimated the treatment rate for hyperlipidemia as a result. However, substantial treatment gaps among individuals with diabetes suggest true missed opportunities for care. Similarly, because antihypertensive therapy is guided by overall cardiovascular risk rather than blood pressure thresholds alone, our approach likely underestimated hypertension treatment rates. Fourth, definitions of cardiometabolic risk factors relied self-reported diagnoses, in addition to biometric and/or laboratory measures. Fifth, because NHANES does not capture subclinical CVD or biomarkers (eg, high-sensitivity troponin or N-terminal pro-B-type natriuretic peptide), our definition of CKM stage 2 and above may have misclassified individuals with biomarker-defined or subclinical cardiovascular disease but without overt clinical disease or CKM risk factors. Last, the absence of data on atrial fibrillation or peripheral artery disease in NHANES likely excluded some adults with stage 4 CKM.

CONCLUSIONS

In this national study of U.S. adults with CKM syndrome, treatment rates for hypertension and hyperlipidemia were approximately 50%, compared with substantially higher rates for diabetes. Gaps in treatment initiation were most pronounced among younger adults, women, and Hispanic adults. Even among those receiving therapy, fewer than one-half achieved adequate blood pressure or glycemic control, with the poorest control observed among individuals at highest CVD risk. Together, these findings highlight critical gaps in both treatment initiation and therapeutic intensification and underscore the urgent need for targeted, equitable strategies to improve both treatment and sustained cardiometabolic risk factor control in this high-risk population.

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APPENDIX For supplemental tables, figures, and methods, please see the online version of this paper.